

HORNSBY GIRLS HIGH SCHOOL



Mathematics

Year 12 Higher School Certificate
Half-Yearly Examination 2013

STUDENT NUMBER: _____

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided
- In Questions 6 – 10, show relevant mathematical reasoning and/or calculations
- Marks may be deducted for untidy and poorly arranged work
- Do not use correction fluid or tape
- Do not remove this paper from the examination

Total marks – 80

Section I Pages 2 – 4

5 marks

Attempt Questions 1 – 5

Answer on the Objective Response Answer Sheet
provided

Section II Pages 5 – 11

75 marks

Attempt Questions 6 – 10. Start each question in a new writing booklet. Write your student number and fill in required information on each writing booklet.

At the end of the assessment:

- Order you solutions, starting with Objective Response answer sheet, then Questions 6 – 10
- Place the question paper on top.
- Do NOT staple

Question	1-5	6	7	8	9	10	Total
Total	/5	/15	/15	/15	/15	/15	/80

This assessment task constitutes 25% of the Higher School Certificate Course School Assessment

Section I

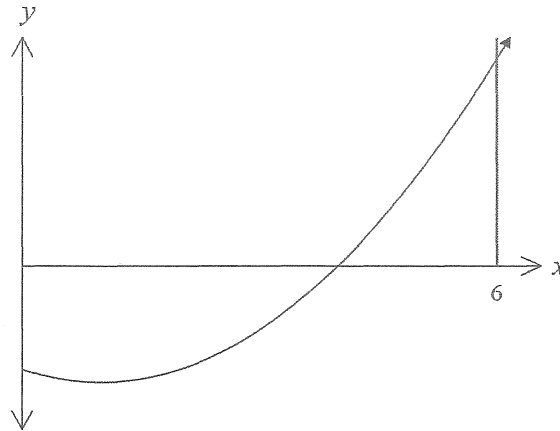
5 marks

Attempt Questions 1 – 5

Allow about 10 minutes for this section

Use the Objective Response answer sheet for Questions 1 – 5

- 1 The diagram below shows the graph of $y = x^2 - 2x - 8$.



What is the correct expression for the area bounded by the x -axis and the curve $y = x^2 - 2x - 8$ between $0 \leq x \leq 6$?

- (A) $A = \int_0^5 (x^2 - 2x - 8)dx + \left| \int_5^6 (x^2 - 2x - 8)dx \right|$
- (B) $A = \int_0^4 (x^2 - 2x - 8)dx + \left| \int_4^6 (x^2 - 2x - 8)dx \right|$
- (C) $A = \left| \int_0^5 (x^2 - 2x - 8)dx \right| + \int_5^6 (x^2 - 2x - 8)dx$
- (D) $A = \left| \int_0^4 (x^2 - 2x - 8)dx \right| + \int_4^6 (x^2 - 2x - 8)dx$
- 2 What is the solution to the equation $\log_e(x+2) - \log_e x = \log_e 4$?

- (A) $\frac{2}{5}$
- (B) $\frac{2}{3}$
- (C) $\frac{3}{2}$
- (D) $\frac{5}{2}$

3 What is the derivative of $\log_2 x$?

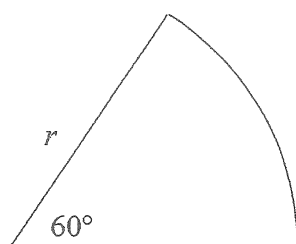
(A) $\frac{1}{x}$

(B) $\frac{1}{2x}$

(C) $\ln 2x$

(D) $\frac{1}{x \ln 2}$

4 The sector below has an area of 10π square units.



Not to scale

What is the value of r ?

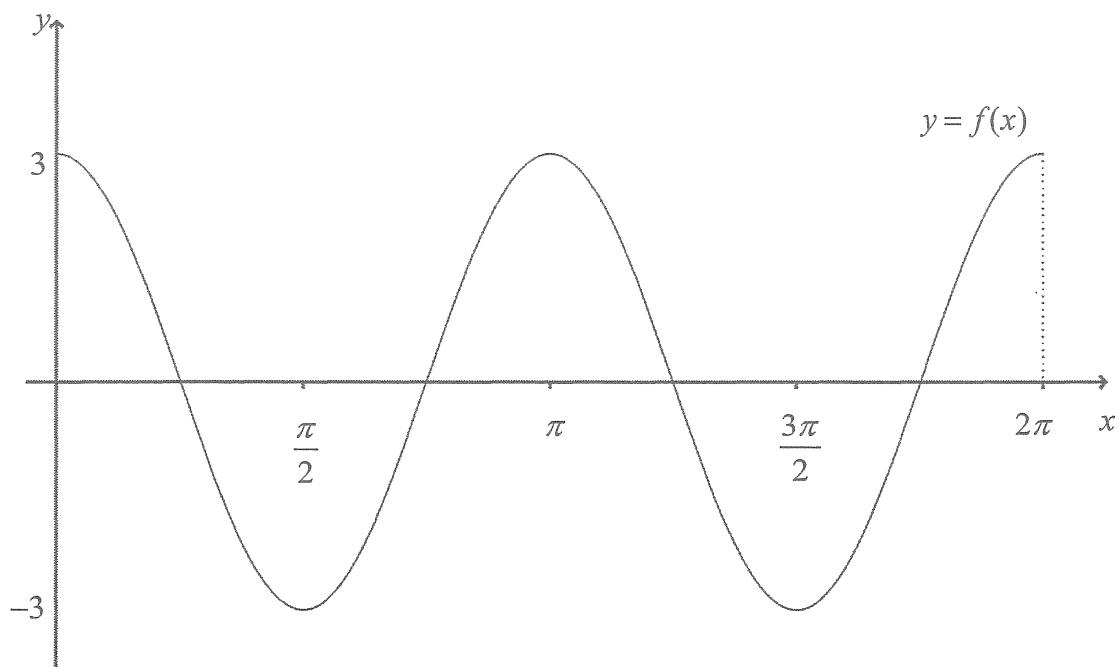
(A) $\sqrt{60}$

(B) $\sqrt{60\pi}$

(C) $\sqrt{\frac{\pi}{3}}$

(D) $\sqrt{\frac{1}{3}}$

- 5 The diagram shows a sketch of the graph $y = f(x)$, for $0 \leq x \leq 2\pi$.



The function $f(x)$ is:

- (A) $f(x) = 3\cos 2x$
- (B) $f(x) = 2\cos 3x$
- (C) $f(x) = 3\cos \frac{x}{2}$
- (D) $f(x) = 2\cos \frac{x}{3}$

End of Section I

Section II

75 marks

Attempt Questions 6 – 10

Allow about 1 hour and 50 minutes for this section

Begin each question in a new writing booklet, indicating the question number.

Extra writing booklets are available.

Question 6 (15 marks) Start a new writing booklet.

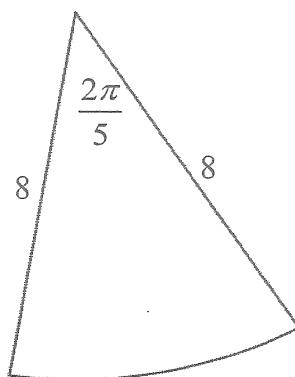
- (a) Solve $5^{4x-3} = 1$ 2
- (b) Find $\log_7 5$ correct to 2 decimal places. 2
- (c) Simplify $3\log 5 - \log 25$, leaving your answer in exact form. 2
- (d) Solve $\log_3 x = 4$ 1
- (e) (i) Sketch the curve $y = \log_{10}(x+1)$ 2
(ii) State the domain of the function in (i). 1
- (f) Evaluate e^{-5} correct to 2 significant figures. 2
- (g) Differentiate $y = e^{2x}$. 1
- (h) Find the equation of the tangent to the curve $y = 2e^x$ at the point where $x = -1$. 2

Question 7 (15 marks) Start a new writing booklet.

- (a) Convert $\frac{7\pi}{9}$ radians to degrees.

1

(b)



The diagram above shows a sector of a circle. Find the perimeter of the circle, correct to one decimal place.

2

- (c) Find $\tan(2.4)$ correct to two decimal places.

1

- (d) Write down the exact value of $\tan\frac{11\pi}{6}$.

2

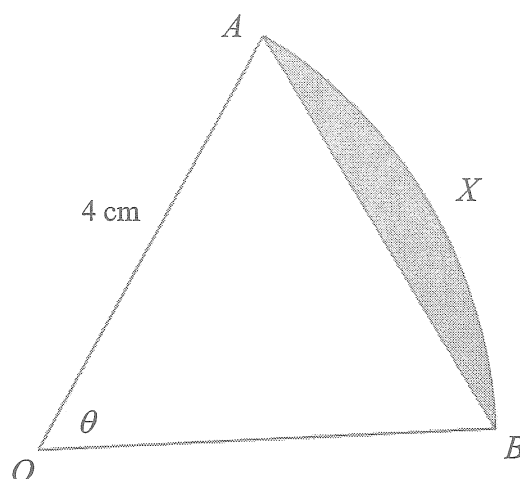
- (e) Solve $2\sin\theta - \sqrt{3} = 0$ for $0 \leq \theta \leq 2\pi$.

2

Question 6 continues on page 7

Question 7 (continued)

(f)



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The area of the sector shown above is $8\pi \text{ cm}^2$.

(i) Find θ in radians. 2

(ii) Find the exact area of the shaded segment. 2

(g) For the function $y = 5 \sin \frac{x}{2}$,

(i) State the amplitude. 1

(ii) State the period. 1

(iii) Sketch its graph in the domain $0 \leq x \leq 2\pi$. 1

End of Question 7

Question 8 (15 marks) Start a new writing booklet

(a) Differentiate the following with respect to x :

(i) $y = \log_e \sqrt{2x+1}.$ 2

(ii) $y = 3xe^{4x+1}.$ 2

(iii) $y = \frac{\ln 3x}{x^3}.$ 2

(b) (i) Write $\log_e \frac{3x+2}{2x-1}$ in an equivalent form using logarithm laws. 1

(ii) Hence, or otherwise, find the derivative of $y = \log_e \frac{3x+2}{2x-1}$ in simplest form 2

(c) For the function $f(x) = e^{2x}(2-x)$:

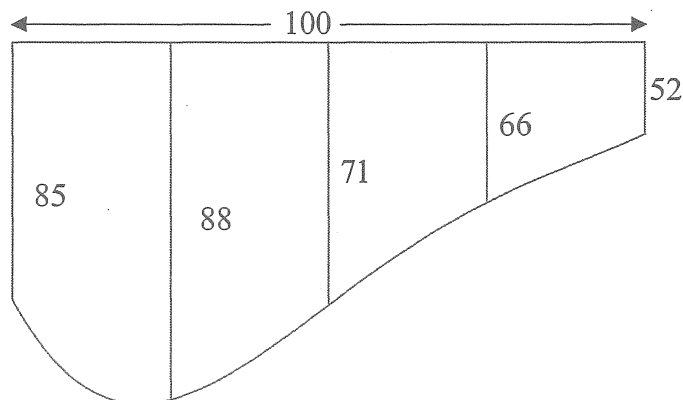
(i) Show that $f'(x) = e^{2x}(3-2x)$ 1

(ii) Find the stationary point of the curve and determine its nature. 3

(d) The diagram shows a paddock, bounded on one side by a river. 2

Use the given measurements and Simpson's rule to approximate the area of the paddock.

Give the answer to the nearest m^2 .



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Question 9 (15 marks) Start a new booklet.

(a) Differentiate the following with respect to x :

(i) $\tan(1-2x)$ 2

(ii) $\sin 3x^2$ 2

(iii) $\frac{x}{\cos x}$ 2

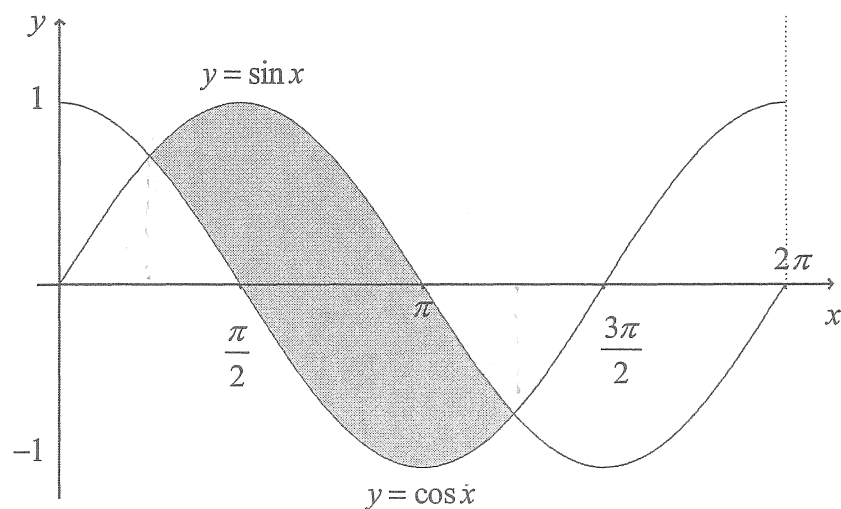
(b) Find:

(i) $\int \cos \frac{x}{3} dx$ 1

(ii) $\int_0^{\frac{\pi}{2}} 4 \sin 2x dx$ 3

(c) The gradient function of a curve $y = f(x)$ is given by $f'(x) = 4 \cos 4x$. 2
Find the equation of the curve if it passes through the point $(0, -2)$.

(d) The diagram below shows the curves $y = \cos x$ and $y = \sin x$.



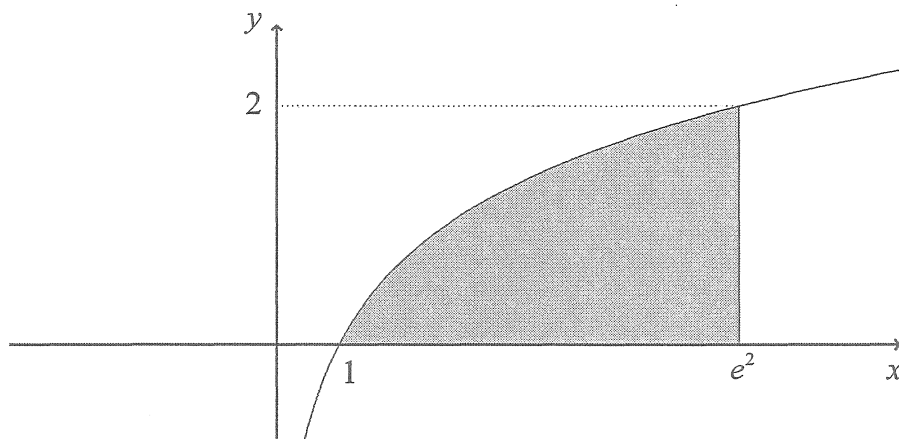
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(i) Show that the curves intersect at $x = \frac{\pi}{4}$ and $x = \frac{5\pi}{4}$. 1

(ii) Find the shaded area. 2

Question 10 (15 marks) Start a new writing booklet.

- (a) (i) Differentiate $y = x \cos x$ 1
- (ii) Hence, or otherwise, find $\int x \sin x dx$. 2
- (b) Evaluate:
- (i) $\int_0^1 \frac{x}{x^2 + 1} dx$. 2
- (ii) $\int_{-1}^3 \frac{-x}{e^{\frac{x}{2}}} dx$. 2
- (c) By considering the area against the y -axis, find the area enclosed by the graph of $y = \log_e x$, the x -axis and $x = e^2$. 3



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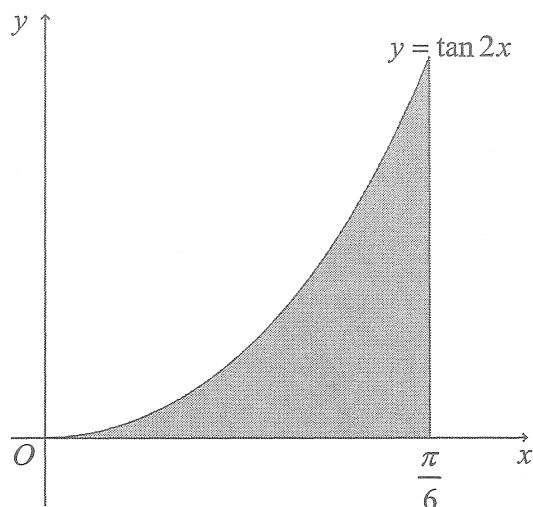
Question 10 continues on page 11

Question 10 (continued)

- (d) (i) By expressing $\sec\theta$ and $\tan\theta$ in terms of $\sin\theta$ and $\cos\theta$, show that $\sec^2\theta - \tan^2\theta = 1$.

1

(ii)



The diagram shows part of the function $y = \tan 2x$. The shaded region is bounded by the curve, the x -axis, and the line $x = \frac{\pi}{6}$. The region is rotated about the x -axis to form a solid.

- (α) Show that the volume of the solid is given by $V = \pi \int_0^{\frac{\pi}{6}} (\sec^2 2x - 1) dx$

2

You may use your result from part (i).

- (β) Find the exact volume of the solid.

2

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Section 1

1. $x^2 - 2x - 8 = 0$ $\frac{x-4}{x+2}$
 $(x-4)(x+2) = 0$
 $\therefore x = 4, -2$

$\therefore$ (D)

2. $\log_e \left(\frac{x+2}{x} \right) = \log_e 4$

$\therefore \frac{x+2}{x} = 4$

$x+2 = 4x$

$2 = 3x$

$\therefore x = \frac{2}{3}$ (B)

3. $\frac{\log_e 2}{\log_e 2} = \frac{1}{\log_e 2} \times \frac{1}{x}$

(D)

4. $A = \frac{1}{2} r^2 \theta$

$10\pi = \frac{1}{2} r^2 \times \frac{\pi}{3}$

$60 = r^2$

$\therefore r = \sqrt{60}$ (A)

5. $a = 3$, Period $= \pi$
 $\therefore n = 2$

$\therefore f(x) = 3 \cos 2x$

(A)

6 a) $5^{4x-3} = 5^0$

$\therefore 4x-3 = 0$

$4x = 3$

$x = \frac{3}{4}$

b) $\log_7 5 = \frac{\log_e 5}{\log_e 7}$

$= \frac{\log_e 5}{\log_e 7}$

$= 0.827087...$

$= 0.83$

c) $3 \log 5 - \log 25 = \log 5^3 - \log 25$

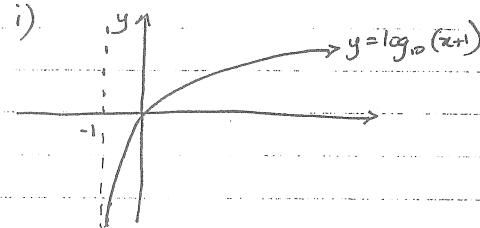
$= \log \frac{125}{25}$

$= \log 5$

d) $3^4 = x$

$\therefore x = 81$

e) i)



ii) domain: $x > -1$

f) $e^{-5} = 6.7379... \times 10^{-3}$
 $= 6.74 \times 10^{-3}$

g) $y' = 2e^{2x}$

h) $y = 2e^x$
 $y' = 2e^x$

when $x = -1$, $y = 2e^{-1}$
 $y' = 2e^{-1}$

$\therefore y - \frac{2}{e} = \frac{2}{e}(x - -1)$

$ey - 2 = 2x + 2$

$\therefore 0 = 2x - ey + 4$

or $(y = \frac{2}{e}x + \frac{4}{e})$

7a) $\frac{7\pi}{9} \times \frac{180}{\pi} = 140^\circ$

b) $p = r\theta + 8 + 8$

$= 8 \times \frac{2\pi}{5} + 16$

$= 26.05309...$

$= 26.1 \text{ u.}$

c) $\tan(2.4)$

$= -0.9160142...$

$= -0.92$ (2 dp)

d) $\tan \frac{11\pi}{6}$ $\frac{s}{c}$

$= -\tan(2\pi - \frac{\pi}{6})$

$= -\tan \frac{\pi}{6}$

$= -\frac{1}{\sqrt{3}}$

e) $2 \sin \theta = \sqrt{3}$
 $\sin \theta = \frac{\sqrt{3}}{2}$

$\therefore \theta = \frac{\pi}{3}, \pi - \frac{\pi}{3}$

$= \frac{\pi}{3}, \frac{2\pi}{3}$

f) i) $A = \frac{1}{2} r^2 \theta$

$2\pi = \frac{1}{2} \times 4^2 \theta$

$2\pi = 8\theta$

$\therefore \theta = \frac{\pi}{4}$

ii) $A = \frac{1}{2} r^2 (\theta - \sin \theta)$

$= \frac{1}{2} \times 4^2 \left(\frac{\pi}{4} - \sin \frac{\pi}{4} \right)$

$= 8 \left(\frac{\pi}{4} - \frac{1}{\sqrt{2}} \right)$

$= 8 \left(\frac{\pi}{4} - \frac{\sqrt{2}}{2} \right)$

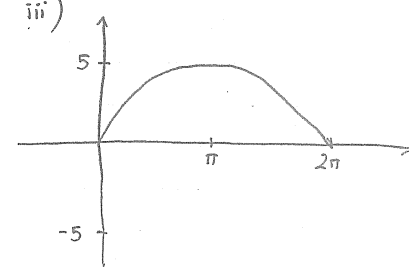
$= 2\pi - 4\sqrt{2} \text{ u}^2$

g) i) $a = 5$

ii) $T = \frac{2\pi}{\frac{1}{2}}$

$= 4\pi$

iii)



s	A
T	C

$$8a) i) y = \log_e (2x+1)^{\frac{1}{2}}$$

$$\therefore y = \frac{1}{2} \log_e (2x+1)$$

$$y' = \frac{1}{2} \times \frac{2}{2x+1}$$

$$= \frac{1}{2x+1}$$

$$ii) y = 3x e^{4x+1}$$

$$y' = 3x e^{4x+1} + 3x \times 4e^{4x+1}$$

$$= 3e^{4x+1} (1+4x)$$

$$iii) y = \frac{\ln 3x}{x^3}$$

$$y' = \frac{x^2 \times \frac{1}{3x} - \ln 3x \times 3x^2}{(x^3)^2}$$

$$= \frac{x^2 - 3x^2 \ln 3x}{x^6}$$

$$= \frac{x^2 (1 - 3 \ln 3x)}{x^6}$$

$$= \frac{1 - 3 \ln 3x}{x^4}$$

$$b) i) \log_e \frac{3x+2}{2x-1} = \log_e (3x+2) - \log_e (2x-1)$$

$$ii) y' = \frac{3}{3x+2} - \frac{2}{2x-1}$$

$$= \frac{3(2x-1) - 2(3x+2)}{(3x+2)(2x-1)}$$

$$= \frac{-7}{(3x+2)(2x-1)}$$

$$c) f(x) = e^{2x} (2-x)$$

$$i) f'(x) = e^{2x} x - 1 + 2e^{2x} (2-x)$$

$$= e^{2x} (-1 + 2(2-x))$$

$$= e^{2x} (3-2x)$$

$$ii) \text{ stat. pt. when } f'(x) = 0$$

$$\text{i.e. } e^{2x} (3-2x) = 0$$

$$3-2x = 0$$

$$3 = 2x$$

$$\therefore x = \frac{3}{2}$$

$$\text{when } x = \frac{3}{2} \quad y = e^{2 \times \frac{3}{2}} \left(2 - \frac{3}{2}\right)$$

$$= \frac{e^3}{2}$$

x	1.4	1.5	1.6
f(x)	3.288	0	-83.4

$\therefore \text{'max.'}$

$\therefore$ maximum turning point at $\left(\frac{3}{2}, \frac{e^3}{2}\right)$

$$d) A = \frac{b-a}{6} \left\{ f(a) + 4f\left(\frac{a+b}{2}\right) + f(b) \right\}$$

0	25	50	75	100
85	88	71	66	52

$$= \frac{50-0}{6} \left\{ 85 + 4 \times 88 + 2 \times 71 + 4 \times 66 + 52 \right\}$$

$$= 7458.3 \text{ m}^2$$

$$\approx 7458 \text{ m}^2$$

$$1a) i) \frac{d}{dx}(\tan(1-2x))$$

$$= -2\sec^2(1-2x)$$

$$ii) \frac{d}{dx}(\sin 3x^2)$$

$$= 6x \cos 3x^2$$

$$iii) \frac{d}{dx} \left(\frac{x}{\cos x} \right)$$

$$= \frac{\cos x \times 1 - x \times -\sin x}{(\cos x)^2}$$

$$= \frac{\cos x + x \sin x}{\cos^2 x}$$

$$b) i) \int \cos \frac{x}{3} dx$$

$$= 3 \sin \frac{x}{3} + C$$

$$ii) \int_0^{\frac{\pi}{2}} 4 \sin 2x dx$$

$$= \left[-\frac{4}{2} \cos 2x \right]_0^{\frac{\pi}{2}}$$

$$= (-2 \cos \frac{2\pi}{2} - -2 \cos 0)$$

$$= 2 + 2$$

$$= 4$$

$$c) f'(x) = 4 \cos 4x$$

$$f(x) = \frac{4}{4} \sin 4x + c$$

$$\text{When } x=0, f(x) = -2$$

$$\text{ie. } -2 = \sin 0 + c$$

$$\therefore c = -2$$

$$\therefore f(x) = \sin 4x - 2$$

$$d) y = \cos x \quad y = \sin x$$

$$i) \therefore \cos x = \sin x$$

$$\therefore 1 = \frac{\sin x}{\cos x}$$

$$\text{ie. } \tan x = 1$$

$$\therefore x = \frac{\pi}{4}, \frac{\pi + \pi}{4}$$

$$= \frac{\pi}{4}, \frac{5\pi}{4}$$

$$ii) A = \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} (\sin x - \cos x) dx$$

$$= \left[-\cos x - \sin x \right]_{\frac{\pi}{4}}^{\frac{5\pi}{4}}$$

$$= \left(-\cos \frac{5\pi}{4} - \sin \frac{5\pi}{4} \right) - \left(-\cos \frac{\pi}{4} - \sin \frac{\pi}{4} \right)$$

$$= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right)$$

$$= \frac{2}{\sqrt{2}} + \frac{2}{\sqrt{2}}$$

$$= \frac{4}{\sqrt{2}} \text{ or } 2\sqrt{2} \text{ u}^2$$

$$10a) i) y = x \cos x$$

$$y' = x \times -\sin x + 1 \times \cos x$$

$$= \cos x - x \sin x$$

$$ii) \int x \sin x dx = - \int (\cos x - x \sin x) dx + \int \cos x dx$$

$$= -x \cos x + \sin x + C$$

$$b) i) \int_0^1 \frac{x}{x^2+1} dx = \frac{1}{2} \int_0^1 \frac{2x}{x^2+1} dx$$

$$= \frac{1}{2} \left[\ln(x^2+1) \right]_0^1$$

$$= \frac{1}{2} [\ln 2 - \ln 1]$$

$$= \frac{1}{2} \ln 2$$

$$ii) \int_{-1}^3 e^{-\frac{x}{2}} dx = \left[-2e^{-\frac{x}{2}} \right]_{-1}^3$$

$$= -2e^{-\frac{3}{2}} - -2e^{-\frac{1}{2}}$$

$$= -2e^{-\frac{3}{2}} + 2e^{-\frac{1}{2}} \quad \text{(or } 2(e^{-\frac{1}{2}} - e^{-\frac{3}{2}}))$$

$$c) \text{ If } y = \log_e x \Rightarrow x = e^y$$

$$\therefore \text{Area against } y\text{-axis : } A = \int_0^2 e^y dy$$

$$= \left[e^y \right]_0^2$$

$$= e^2 - e^0$$

$$= e^2 - 1$$

$$\text{Now enclosed area is rectangle - area against } y\text{-axis}$$

$$\text{ie. } 2e^2 - (e^2 - 1)$$

$$= e^2 + 1 \text{ u}^2$$

$$i) \sec \theta = \frac{1}{\cos \theta} \quad \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\begin{aligned} \therefore \sec^2 \theta - \tan^2 \theta &= \frac{1}{\cos^2 \theta} - \frac{\sin^2 \theta}{\cos^2 \theta} \\ &= \frac{1 - \sin^2 \theta}{\cos^2 \theta} \\ &= \frac{\cos^2 \theta}{\cos^2 \theta} \\ &= 1 \end{aligned}$$

$$ii) a) V = \pi \int_a^b y^2 dx$$

$$y = \tan 2x$$

$$\therefore y^2 = \tan^2 2x$$

$$= \pi \int_0^{\frac{\pi}{6}} (\sec^2 2x - 1) dx$$

$$\text{from (i)} \sec^2 \theta - \tan^2 \theta = 1$$

$$\therefore \tan^2 \theta = \sec^2 \theta - 1$$

hence

$$\tan^2 2x = \sec^2 2x - 1$$

$$b) V = \pi \left[\tan^2 2x - x \right]_0^{\frac{\pi}{6}}$$

$$= \pi \left[\left(\tan^2 \frac{2\pi}{6} - \frac{\pi}{6} \right) - \left(\tan^2 0 - 0 \right) \right]$$

$$= \pi \left[\sqrt{3} - \frac{\pi}{6} \right]$$

$$= \sqrt{3} \pi - \frac{\pi^2}{6} \text{ or } \frac{6\sqrt{3}\pi - \pi^2}{6}$$